

Quantifying Investor Sentiment with Multimodal Data in the Chinese Stock Market

By YI YU*

This study is based on the visual and textual information from the homepage of Sina News. It employs the Vision Transformer model to classify the sentiment of news images and constructs a photo-based pessimism index, PhotoPes. Additionally, a pre-trained language model based on BERT is used to build a text-based pessimism index, TextPes. Empirical analysis over the 2014–2026 sample period reveals that sentiment changes in news images significantly affect market returns, with PhotoPes_{t-3} demonstrating significant negative predictive effects across multiple Chinese stock indices—particularly the ChiNext Composite (1% level) and smaller-cap indices (10% level). Moreover, image and text sentiment indicators exhibit complementary relationships in forecasting stock market returns, providing empirical support for multimodal sentiment quantification. The study further applies these sentiment indicators to construct a rule-based trading strategy that achieves positive risk-adjusted returns, outperforming the CSI 500 benchmark. The findings indicate that integrating image and text sentiment information can effectively enhance market prediction and portfolio performance, offering new perspectives for investor decision-making and market regulation.

JEL: G12, G14, G41

Keywords: Investor sentiment, Behavioral finance, Machine learning

In recent years, investor sentiment, as a significant research direction in behavioral finance, has garnered considerable attention from the academic community due to its widespread impact on market fluctuations and investor behavior. Tetlock (2007) argued that investor sentiment not only directly affects market returns and trading volumes but also has a significant effect on market volatility, especially during periods of market panic or intense fluctuation. Quantifying investor sentiment is a key focus in financial research. In existing studies, scholars have mostly focused on quantifying the sentiment of textual content from sources like news articles and social media data, using Natural Language Processing (NLP) tools to construct text-based sentiment indicators to study the relationship between sentiment changes and market returns. These studies, primarily based on

* Yu: M.S. in Data Science student, New York University (yy5919@nyu.edu); B.A. in Economics, Lingnan College, Sun Yat-sen University (2024). I am grateful to my advisor, Associate Professor Xiaobin Liu (liuxb53@mail.sysu.edu.cn) of Lingnan College, Sun Yat-sen University, for his invaluable guidance. All remaining errors are my own.

data sources from news text, have revealed a significant correlation between negative news and a decline in market returns. However, a research framework that relies solely on textual data has certain limitations and cannot fully capture the complex characteristics of investor sentiment.

With the diversification of information dissemination media, images, as an important component of news, can quickly convey complex information due to their intuitive, vivid, and lively nature, and they demonstrate a stronger effect in evoking emotions than text. Garcia and Stark (1991)’s research, using eye-tracking technology to analyze how readers read newspapers, found that visual elements, represented by images, are typically the first content that readers notice and can attract their attention faster than words. However, the role of images in financial sentiment analysis has not been fully explored, and research on the relationship between image sentiment and financial markets is relatively scarce. Obaid and Pukthuanthong (2022) utilized machine learning techniques to classify the sentiment of news images and constructed a photo pessimism index (PhotoPes) to study the impact of image sentiment on market behavior. This research demonstrated the importance of image sentiment in sentiment quantification studies, but its scope was mainly limited to the European and American market contexts, with little research involving the Chinese stock market.

In the Chinese stock market, the content and form of news media have a significant impact on investor behavior and market performance. The Chinese stock market is composed of major stock indices such as the Shanghai Composite Index, the Shenzhen Component Index, and the CSI 300 Index. The fluctuations of these indices not only reflect changes in market fundamentals but are also largely driven by investor sentiment. Although some domestic research has begun to focus on the quantification of news sentiment, it has mainly concentrated on the analysis of text, with a lack of systematic research on news images, an important source of emotional information. Particularly in a market like China, where the trading activity of retail investors is relatively high, images as direct visual stimuli may have a more direct impact on investment decisions. Therefore, by combining sentiment analysis of news images and text data, this paper explores the impact of image and text information in the news on investor behavior, constructs effective indicators that reflect the emotional fluctuations of the Chinese stock market, which will provide a new perspective for market behavior research and strong support for investor decision-making and market supervision.

This paper combines the analysis of news images and text sentiment to propose and construct a daily market-level investor sentiment index based on news photos and text, used to study the impact of sentiment fluctuations on the Chinese stock market. The main contributions of the research can be summarized as follows:

First, this paper utilizes the Vision Transformer model from deep learning to classify the sentiment of news images, addressing the technical challenges of large-scale image sentiment analysis, and constructs a photo-based pessimism indicator. Through the analysis of this indicator, the study finds that changes in news im-

age sentiment have a significant effect on stock market return reversals, especially during periods of extreme sentiment, where the predictive power of image sentiment is more prominent. Compared to traditional text analysis methods, the image sentiment indicator can more intuitively and vividly capture changes in investor sentiment, demonstrating the unique advantages of image information in the sentiment quantification of financial markets.

Second, this paper uses the BERT model to analyze the sentiment in news texts, constructing a text-based pessimism indicator. By comparing the pessimistic sentiment in news images with that in news texts, the study finds that image sentiment and text sentiment have a complementary relationship in predicting stock market returns. This finding provides important empirical support for multimodal sentiment analysis, further enriching the perspectives and methods of sentiment quantification research.

Furthermore, this paper applies the sentiment indicators to construct a quantitative trading strategy. The strategy demonstrates strong risk-adjusted returns, outperforming its benchmark, though its factor-model alpha is positive but not statistically significant at conventional levels. An out-of-sample analysis using an expanding-window Ridge regression model (lags 3–5 of PhotoPes as predictors) confirms that sentiment predictability extends beyond the estimation sample: the ChiNext Composite achieves $R_{OOS}^2 = 0.29\%$ significant at the 5% level ($t = 2.02$), and the SZSE Composite and CSI 500 are significant at the 10% level, providing evidence that the predictive relationship is not purely an in-sample artifact.

Through these innovative methods and empirical analysis, this paper provides a new perspective for the research on quantifying investor sentiment and offers support for investor decision-making and market supervision.

I. Literature Review

A. Theoretical Foundations of Investor Sentiment

Traditional financial theory, centered on the *Efficient Market Hypothesis (EMH)*, posits that asset prices fully reflect all available information, and market participants are rational actors with complete information-processing capabilities. However, frequent market anomalies, such as price bubbles and excessive volatility, are difficult to explain within this traditional framework. Behavioral finance emerged as a response, integrating psychological insights into financial research and arguing that investors' cognitive biases and emotional fluctuations are significant factors causing markets to deviate from fundamentals.

In the early development of behavioral finance, the *noise trader theory* proposed by De Long et al. (1990) laid the theoretical groundwork for investor sentiment research. This theory suggests the existence of a group of irrational “noise traders” who trade based on sentiment rather than fundamentals, causing asset prices to deviate from their intrinsic values. Crucially, even the presence of rational arbitrageurs cannot fully counteract the influence of noise traders, as

arbitrage activities face numerous constraints. Furthermore, the theory indicates that the additional risk introduced by noise traders—*noise trader risk*—deters arbitrageurs from taking sufficiently large positions, even when they know an asset is mispriced.

Building on noise trader theory, Shleifer and Vishny (1997) systematically elaborated on the *limits to arbitrage* theory, further explaining the mechanisms through which irrational behavior can persistently affect asset prices. They pointed out that the main constraints faced by arbitrageurs include fundamental risk, noise trader risk, and implementation costs. These factors collectively prevent arbitrage activities from immediately eliminating market mispricing. The limits to arbitrage theory provides a key mechanical explanation for how sentiment influences asset prices, namely that the interaction of market frictions and investor sentiment produces lasting price deviations.

Baker and Wurgler (2007) developed a complete analytical framework for the impact of investor sentiment on asset pricing based on the preceding theories. They emphasized that the influence of investor sentiment varies systematically across different assets, having a greater impact on those that are more subjective to value and harder to arbitrage. Such assets typically include new-economy firms, loss-making companies, high-growth firms, and high-volatility stocks. This research not only revealed the intrinsic link between asset characteristics and sentiment sensitivity but also provided a clear direction for subsequent empirical studies.

The theory of investor sentiment provides a systematic framework for explaining irrational market phenomena. It not only clarifies how sentiment factors influence market prices through noise traders but also reveals how the limits to arbitrage mechanism allows these effects to persist, and it identifies the asset characteristics and market conditions under which sentiment has the greatest impact. As research has deepened, these theories have offered new perspectives for understanding market volatility and investment behavior.

B. The Evolution of Sentiment Quantification Methods

While investor sentiment theory provides a framework for understanding market irrationality, accurately quantifying sentiment has been a long-standing challenge. With advancements in research and technology, sentiment quantification methods have evolved significantly, moving from single indicators to multivariate integration and from indirect inference to direct measurement.

Early research primarily used market technical indicators as proxies for sentiment. Lee, Shleifer and Thaler (1991) used the closed-end fund discount as a sentiment indicator, finding that its fluctuations could predict the returns of small-cap stocks. Other commonly used market indicators include trading volume, IPO volume and first-day returns, and option-implied volatility. Although these indicators provide indirect measures of sentiment, they each have limitations and cannot fully capture the multidimensional nature of sentiment.

To overcome the limitations of single indicators, Baker and Wurgler (2006) constructed a composite sentiment index by combining multiple market indicators using *principal component analysis*, significantly improving the comprehensiveness of sentiment measurement. This innovative method was subsequently widely adopted and extended, becoming a major milestone in sentiment quantification.

The application of text analysis techniques marked a major advancement in sentiment quantification. Tetlock (2007), through the textual analysis of a Wall Street Journal column, was the first to systematically demonstrate that media pessimism could predict market performance, opening a new path for sentiment research. Addressing the limited accuracy of general-purpose dictionaries in a financial context, Loughran and McDonald (2011) developed a specialized financial dictionary, which substantially improved the accuracy of sentiment analysis in financial texts and laid the foundation for subsequent research.

With the rise of the internet, the sources of sentiment data further expanded. Bollen, Mao and Zeng (2011) analyzed public mood using Twitter data, finding that changes in specific mood dimensions could predict the Dow Jones Industrial Average. Similarly, Chen et al. (2014) analyzed investor opinions on the Seeking Alpha website, demonstrating that content from such specialized social media platforms contains valuable predictive information. These studies not only expanded the sources of sentiment data but also improved the real-time availability and granularity of sentiment measurement. Concurrently, with the increased availability of search engine data, investor attention became an important dimension of sentiment research. Da, Engelberg and Gao (2011) used the Google *Search Volume Index (SVI)* to measure investor attention, finding that it could predict short-term price pressure and long-term return reversals.

Text analysis methods also evolved from simple word frequency counts to complex machine learning models. As machine learning technology advanced, methods such as *Support Vector Machine (SVM)* and *Decision Trees* were widely applied to sentiment analysis. Pang, Lee et al. (2008) systematically compared the performance of different machine learning algorithms in text sentiment classification, demonstrating that SVM has a distinct advantage in handling high-dimensional feature spaces. Similarly, Antweiler and Frank (2004) applied machine learning methods to analyze information from online forums, successfully predicting market volatility and laying the groundwork for computer-assisted sentiment analysis.

C. Visual Information and Investor Sentiment

Research in psychology and neuroscience provides strong support for the unique advantages of visual information in evoking and conveying emotions. Through mechanisms such as dual-system thinking and risk-as-feelings, this advantage significantly impacts investment decisions. Although research on visual sentiment in finance is still in its early stages, existing evidence suggests that images in financial news contain valuable emotional information that can systematically

influence market performance. This research direction holds vast potential for development and significant practical importance.

Psychological studies show that visual information processing is characterized by speed, directness, and emotional intensification. Powell et al. (2015) confirmed through comparative experiments that, compared to purely textual information, image stimuli can elicit stronger and more lasting emotional responses, which are often independent of rational cognitive processes. The constructivist view proposed by Lindquist et al. (2012) suggests that emotional experiences are generated through the synergistic action of multiple basic psychological operations. Visual stimuli can act as inputs to trigger this construction process, but the formation of emotion involves the joint participation of both automatic and controlled cognitive processes.

These characteristics of visual information processing align closely with the *Dual-System Theory* proposed by Kahneman (2011). According to this theory, human thinking involves two systems: System 1 (fast, automatic, and emotionally driven) and System 2 (slow, rational, and effortful). Visual information is primarily processed by System 1, thus enabling it to rapidly induce emotional reactions, whereas textual information relies more on the rational analysis of System 2. This difference is particularly important in the financial market environment, where investors in a fast-paced market often rely on System 1 to make decisions.

The *risk-as-feelings hypothesis* by Loewenstein et al. (2001) further reinforces the importance of visual information in investment decisions. This hypothesis posits that when faced with risky decisions, people often make judgments based on immediate emotional responses rather than rational calculations. Because visual information has a strong capacity to elicit emotions, it can significantly influence investors' risk perception and subsequent decision-making, especially in highly uncertain market environments.

Although the emotional effects of visual information have been extensively studied in psychology, its application in the financial domain is still an emerging field. Early financial research focused mainly on the informational content of charts and pictures in corporate annual reports, with even more limited research on the impact of images in financial media.

D. The Application of Deep Learning in Financial Sentiment Analysis

The rapid development of deep learning technology has brought transformative changes to financial sentiment analysis, greatly enhancing the accuracy and efficiency of processing unstructured data. This section will systematically explore the progress of deep learning applications in financial sentiment analysis, with a particular focus on breakthroughs in text analysis, image analysis, and multimodal fusion.

In the field of text sentiment analysis, deep learning methods have shown significant advantages over traditional approaches. Peng and Jiang (2015) were among the first to apply word embeddings and deep neural networks to finan-

cial news analysis, predicting stock price movements by integrating news text information and significantly improving prediction accuracy compared to traditional methods that rely solely on historical price data. Subsequently, Kraus and Feuerriegel (2017) combined Long Short-Term Memory (*LSTM*) networks with transfer learning to analyze financial disclosure documents, demonstrating the superiority of deep learning methods in capturing long-term dependencies in text and predicting market reactions, while also addressing the issue of insufficient labeled data in the financial domain.

The emergence of pre-trained language models marked a major breakthrough in text sentiment analysis. The BERT (*Bidirectional Encoder Representations from Transformers*) model proposed by Devlin et al. (2018) achieved groundbreaking performance on multiple natural language processing tasks through large-scale unsupervised pre-training and a bidirectional context encoding mechanism. In its application to finance, Yang, Uy and Huang (2020) developed the FinBERT model based on BERT, which was pre-trained on financial texts, further improving the accuracy and domain adaptability of financial text sentiment analysis.

Compared to text analysis, image sentiment analysis in finance started later but has developed rapidly. The introduction of deep learning has significantly enhanced the capabilities of image sentiment analysis. You et al. (2015) were the first to apply a *Convolutional Neural Network (CNN)* to large-scale image sentiment classification, achieving a substantial increase in accuracy. In a specific application to financial image analysis, Obaid and Pukthuanthong (2022) used the Google Inception (v3) CNN model to analyze the sentiment of financial news images, demonstrating the effectiveness and applicability of deep learning in this specialized field.

In recent years, the emergence of the Vision Transformer (*ViT*) has marked a new phase in image analysis technology. The ViT model, proposed by Dosovitskiy et al. (2020), abandoned the convolutional operations of traditional CNNs and directly applied the Transformer architecture to image processing, showing excellent performance in capturing global information and long-range dependencies. Although the application of ViT in financial image sentiment analysis is still in its infancy, its powerful expressive capabilities and flexibility signal great potential for future applications.

Multimodal fusion is an important direction in financial sentiment analysis, aiming to simultaneously analyze and integrate information from different modalities to obtain a more comprehensive sentiment representation. The research by Obaid and Pukthuanthong (2022) is a significant attempt at multimodal sentiment analysis in finance. They not only analyzed image sentiment but also studied its relationship with text sentiment, finding that the two have a substitute, rather than complementary, relationship in predicting market returns. This multimodal analysis approach offers a new perspective for a comprehensive understanding of market sentiment.

Although the application of deep learning in financial sentiment analysis is

promising, it still faces several challenges, such as a lack of model interpretability, high costs of data quality and annotation, and domain specificity. However, with continuous technological advancements, these challenges are gradually being addressed. Deep learning has brought revolutionary changes to financial sentiment analysis, from pre-trained language models for text processing to the Vision Transformer for image analysis, and multimodal fusion techniques, all of which have greatly enhanced the precision and scope of financial sentiment quantification.

E. Research on Investor Sentiment in the Chinese Stock Market

As the world’s second-largest stock market, China’s unique market mechanisms, investor structure, and information environment suggest that sentiment factors may play a more critical role. This section will systematically review the main progress of research on investor sentiment in the Chinese stock market, explore sentiment quantification methods with Chinese characteristics, and analyze the differential impact of sentiment factors.

Retail investors are more active in the Chinese stock market compared to other markets, a characteristic that makes the market more susceptible to sentiment. The research by Kumar and Lee (2006) shows that in markets with a high proportion of retail investors, the impact of investor sentiment on stock returns is more significant, and the trading behavior of retail investors often exhibits stronger co-movement, thereby amplifying the impact of sentiment shocks on the market. This influence is not only reflected in returns; Foucault, Sraer and Thesmar (2011) further confirmed a significant positive correlation between the trading activity of retail investors and market volatility, indicating that the emotional fluctuations of retail investors may be an important source of excessive market volatility. However, the impact of retail sentiment differs significantly from that of institutional investors. Research by Liu and Liu (2014) on the Shanghai Stock Exchange A-share market revealed this: institutional investor sentiment has predictive power for future market performance, whereas individual investor sentiment does not, indicating a fundamental difference in the quality of sentiment between different types of investors.

An important innovation in Chinese market sentiment research is the use of data from local internet platforms. Meng, Meng and Hu (2016) combined text mining techniques and the *Baidu Index* to construct an investor sentiment index by analyzing information from CSSCI journals on CNKI and Sina Weibo topics. Bu et al. (2018) used posts from Eastmoney’s Guba forums, employing a Naive Bayes method to construct a sentiment indicator that incorporates both bullish/bearish expectations from stock reviews and investor attention. The study found that while overall investor sentiment had no predictive power for the stock market, sentiment from stock reviews during non-trading hours before the market open could predict the opening price, providing new evidence for understanding the behavior of Chinese retail investors. Similarly, Wen et al. (2019) used the Baidu

Search Index as a proxy for retail investor attention, revealing a positive correlation between investor attention and future stock price crash risk, a relationship that is particularly pronounced in firms with high information opacity and low analyst coverage.

The media plays a unique role in the transmission of sentiment in the Chinese market. Wang and Wu (2015) constructed a tone indicator for media reports on IPO companies before their listing and found a significant negative correlation between negative media tone and the IPO underpricing rate. This suggests that media reports directly influence investor sentiment, which in turn affects market pricing efficiency. They also revealed that issuing companies and underwriters intentionally use media promotion to guide investor sentiment, demonstrating the special function of the media as a tool for sentiment guidance. Similarly, Zhang and Wu (2021), using machine learning-based text analysis, studied the contagion effect of media sentiment. They found that optimistic media sentiment significantly influences the optimistic bias of analyst forecasts and is transmitted to the market through two channels: "limited analyst attention" and "investor sentiment," ultimately affecting stock price volatility and tail risk. These findings collectively confirm the key role of the media in the formation and transmission of sentiment in the Chinese market.

Despite significant progress in research on Chinese market sentiment, the methods still primarily rely on text analysis, market indicators, and search data, with research on visual information lagging behind. Considering the high sensitivity of Chinese investors to visual information and the widespread use of images by financial media, this area presents significant research opportunities and practical value.

F. Research Review and Outlook

Existing research on sentiment has achieved fruitful results, theoretically clarifying the mechanisms through which sentiment affects the market, methodologically developing a diverse set of tools for sentiment quantification, and empirically verifying the systematic impact of sentiment factors on asset prices. However, this literature review also reveals some clear research gaps and methodological limitations.

Sentiment quantification methods remain predominantly unimodal, with text analysis being the dominant approach, while research on visual information is relatively insufficient. The study by Obaid and Pukthuanthong (2022) began to fill this gap, but it was based solely on U.S. market data, and the sentiment quantification techniques used, Google Inception (v3) and Stanford's CoreNLP, now have more advanced alternatives. The mechanism of visual sentiment in different market environments still requires in-depth investigation.

Although research on sentiment in the Chinese market has made progress, the methods have largely borrowed from international experience, lacking innovative approaches that consider the unique characteristics of the Chinese market and its

investors. In particular, the analysis of visual investor sentiment in the Chinese market is almost non-existent. Given the characteristics of Chinese investors and the media environment, this research direction holds significant theoretical and practical importance.

In response to the aforementioned research gaps and limitations, this paper proposes a multimodal sentiment analysis framework based on deep learning and applies it to the study of the Chinese stock market. The main theoretical contributions and innovations of this paper are as follows:

First, this paper will enrich the research on visual sentiment in the Chinese stock market. By crawling the homepage of Sina News and analyzing the sentiment of news images, it will explore their role in predicting the Chinese stock market. Considering the unique investor structure and media environment in China, this research will provide an important cross-cultural comparative perspective, enriching the application of sentiment theory in different market contexts.

Second, this paper employs the latest deep learning technologies for sentiment analysis, using the BERT model for text processing and the Vision Transformer architecture for image analysis. This will significantly enhance the accuracy and comprehensiveness of sentiment quantification and provide a basis for methodological comparative studies.

Third, this paper explores the relationship between text sentiment and image sentiment, analyzing the relative importance of these two sentiment sources in different market environments and for different types of stocks. This will enrich the theoretical understanding of multimodal sentiment analysis and offer new ideas for the construction of sentiment indicators.

Fourth, this paper investigates the differences in the predictive power of sentiment indicators across various Chinese market indices. This will contribute to a deeper understanding of the mechanisms through which sentiment factors affect stocks with different characteristics, providing insights for the design of differentiated investment strategies.

II. Data and Indicator Construction

This section will introduce the construction methods for the news photo and text pessimism indicators, PhotoPes and TextPes, and provide a statistical analysis and description of the data used.

A. Image Classification

The Vision Transformer (ViT) is a deep learning model that applies the transformer architecture, widely popular in natural language processing, to computer vision tasks. It was first proposed by Dosovitskiy et al. (2020). Unlike traditional convolutional neural networks (CNNs), ViT first divides an input image into fixed-size patches, then linearly maps these patches into a sequence of tokens. These tokens are processed by a transformer encoder and then used for downstream vision tasks. Figure 1 illustrates the overall architecture.

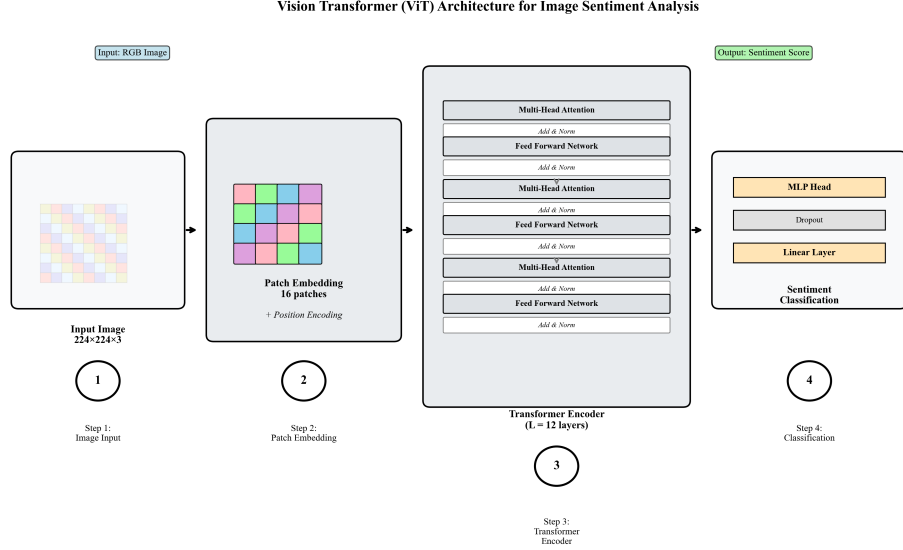


FIGURE 1. DETAILED ViT ARCHITECTURE

Compared to CNNs, ViT has the following advantages: First, the self-attention mechanism of the transformer allows the model to globally model the relationships between image features, whereas CNNs, due to the locality of convolution operations, need to be stacked in multiple layers to obtain a large receptive field. Second, the self-attention weights of ViT provide good interpretability, visually demonstrating which areas of the image the model is focusing on. Third, ViT has stronger scalability, performs excellently when pre-trained on large-scale datasets, and the pre-trained models have good transferability.

This paper performs transfer learning based on ViT, using the image set¹ provided by You et al. (2015) as training data. This dataset contains multiple images, each voted on by five people to determine its sentiment category as positive or negative. To ensure data quality, this paper selected images that all voters agreed were either positive or negative, totaling 882 images, which were used for model training. During the training process, ViT was used as the base model, and hyperparameters were optimized to enhance the model's stability and generalization ability. A low learning rate (5e-6) and high weight decay (0.02) were set to reduce overfitting. The ReduceLROnPlateau scheduling strategy was used during training, where the learning rate was halved if the validation loss did not improve for 3 consecutive epochs, and training was terminated early if there was no improvement for 8 epochs, to avoid wasting computational resources.

Furthermore, to enhance the model's robustness, this paper used cross-entropy

¹The image set can be downloaded from: <https://qzyou.github.io/projects/sa-ds/>

loss with label smoothing and applied various data augmentation strategies, including random cropping, horizontal flipping, rotation, and affine transformations of the images, to improve the model’s adaptability to different inputs. The dataset was split into 70% for the training set, 10% for the validation set, and 20% for the test set to ensure a robust evaluation of the model.

Figure 2 shows the trend of accuracy changes for the training and validation sets during the model’s training process. During training, the model’s training accuracy rapidly increased and stabilized after about 5 epochs, while the validation accuracy fluctuated around 90%. Ultimately, this paper achieved an accuracy of 88.14% on the test set of 177 held-out images, indicating that the model has strong generalization ability on unseen data. The specific evaluation results for the test set are shown in Table 1. The accuracy of the model is calculated as follows:

(1)

$$\text{Accuracy} = \frac{\text{True Positive} + \text{True Negative}}{\text{True Positive} + \text{True Negative} + \text{False Positive} + \text{False Negative}}$$

where True Positive and True Negative represent correctly classified positive and negative image samples, respectively, while False Positive and False Negative represent misclassified image samples.

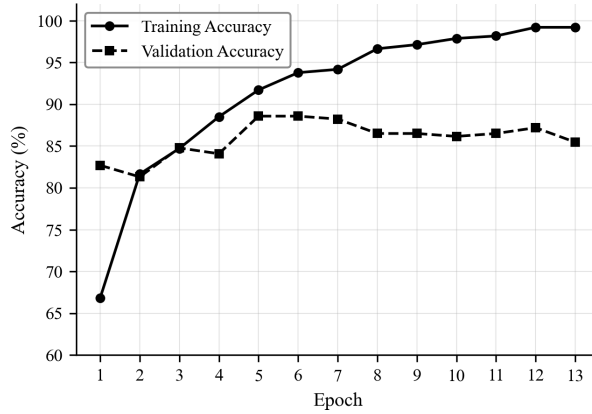


FIGURE 2. TRAINING AND VALIDATION ACCURACY TRENDS

In the appendix, Figure A1 displays the top 10 photos from our sample with the highest predicted probabilities of showing negative ($\text{PhotoNeg}_{it} \text{ max}$) and positive ($\text{PhotoNeg}_{it} \text{ min}$) sentiment. PhotoNeg_{it} refers to the probability that photo i displays negative sentiment on day t . The binary indicator Neg_{it} used in equation (2) is defined as $\mathbf{1}\{\text{PhotoNeg}_{it} > 0.5\}$.

TABLE 1—TEST SET CLASSIFICATION RESULTS

Actual Class	Predicted Positive	Predicted Negative
Positive	115	7
Negative	14	41

B. Image Sample

The data for this study comes from the Sina News website, covering various news categories from 2014 to 2026. This paper primarily focuses on finance-related news, including domestic and international financial markets, the stock market, and finance-related reports in sectors such as lifestyle, technology, real estate, and automobiles. Headline news and general domestic news were also collected to obtain more comprehensive market sentiment information. The news content was collected in two sessions, morning and afternoon, each day to ensure the timeliness and continuity of the data. Through this diversified news source collection strategy, this paper aims to comprehensively capture changes in market sentiment, especially the fluctuations in investor sentiment related to financial markets. The total number of trading days is 2,982, with a total of approximately 245,000 articles collected, averaging approximately 82 articles per day.

C. Variable Construction

The main variable in this paper, PhotoPes, is derived by calculating the proportion of photos predicted to have negative sentiment on a given day. The formula for PhotoPes on day t is as follows:

$$(2) \quad \text{PhotoPes}_t = \frac{\sum_i (\text{Neg}_{it})}{n_t}$$

where Neg_{it} is an indicator variable that indicates whether photo i is predicted to have negative sentiment on day t . The denominator n_t represents the number of photos on day t . PhotoPes is not a simple binary indicator. Although each photo is classified as either having negative or positive sentiment, PhotoPes is a continuous measure. On days with more photos having negative sentiment, the PhotoPes value is higher, while on days with fewer negative sentiment photos, the PhotoPes value is lower.

One of the goals of this research is to explore the relationship between the pessimistic sentiment embedded in images and texts. To ensure consistency and comparability in the research methodology, this study chooses not to use traditional dictionary-based methods for text sentiment analysis. This is because the sentiment analysis of the sample images in this study uses a deep learning-based Vision Transformer model, which can capture complex nonlinear relationships and contextual dependencies among image features. If text analysis were to use a

simple dictionary matching method, it would lead to an asymmetry in the complexity and capability of the sentiment extraction methods for the two media forms, potentially introducing methodological bias and affecting the reliability of the research conclusions.

To overcome the limitations of dictionary-based text sentiment analysis, BERT (Bidirectional Encoder Representations from Transformers), a major breakthrough in the field of natural language processing, provides more powerful technical support for text sentiment analysis. Proposed by the Google AI team in 2018, BERT’s core innovation is its use of a bidirectional context encoding mechanism, which can simultaneously consider the left and right context of words in a sentence, thus more accurately capturing semantic and emotional expressions. Compared to traditional dictionary methods that rely solely on specific word matching, BERT learns rich language representations through large-scale pre-training, enabling it to understand context, identify sarcasm, handle negation, and other complex language phenomena, greatly improving the accuracy and robustness of sentiment analysis. This paper adopts the Chinese pre-trained language model Erlangshen-Roberta-110M-Sentiment² proposed by Zhang et al. (2022) to assess pessimistic sentiment in texts. This model is based on the Chinese RoBERTa-wwm-ext-base architecture proposed by Cui et al. (2020), which is a pre-trained model using a Whole Word Masking strategy, developed by the HFL (Joint Laboratory of Harbin Institute of Technology and iFlytek)³. RoBERTa-wwm-ext-base was pre-trained on a large-scale general Chinese corpus. Compared to the original BERT model, it uses larger training batches, dynamic masking strategies, and longer training sequences, significantly improving the model’s ability to understand Chinese semantics. Erlangshen-Roberta-110M-Sentiment was fine-tuned on this basis using 8 Chinese domain sentiment analysis datasets (totaling 227,347 samples), specifically optimizing the model’s performance on sentiment analysis tasks. This model has shown excellent performance in several sentiment analysis benchmarks. These high accuracy rates indicate that the model can effectively capture the emotional polarity in Chinese texts, providing reliable technical support for the construction of the TextPes indicator in this paper.

By adopting a text sentiment analysis method that matches the complexity of the image analysis method, this paper ensures the comparability of the PhotoPes and TextPes indicators on a methodological basis, providing a more reliable empirical foundation for studying the differences in emotional expression in different media forms and their impact on the market.

The formula for TextPes is as follows:

$$(3) \quad \text{TextPes}_t = \frac{\sum_i (\text{TextNeg}_{it})}{n_t}$$

where TextNeg_{it} is the pessimistic sentiment score for each article i on day t . The

²For details, see: <https://huggingface.co/IDEA-CCNL/Erlangshen-Roberta-110M-Sentiment>

³For details, see: <https://huggingface.co/hfl/chinese-roberta-wwm-ext>

denominator n_t represents the number of articles on day t .

D. Descriptive Statistics

Table 2 presents descriptive statistics for the news sentiment pessimism indicators and major Chinese stock indices for the sample period of 2014–2026, which covers 2,982 trading days. Panel A shows that the mean value of PhotoPes is 0.1900, indicating that approximately 19.0% of news images are predicted to have negative sentiment; the mean value of TextPes is 0.1430, indicating that about 14.3% of news text content is identified as having negative sentiment. In Panel B, the average daily returns of all major Chinese stock market indices are close to zero, which is consistent with the general characteristics of financial markets. From the standard deviation, it can be seen that the ChiNext Composite index has the highest volatility (0.0194), while the SSE Composite Index has the lowest (0.0126), reflecting the different risk characteristics of various market sectors. News data was available to construct the sentiment indicators for all trading days in the sample, with approximately 245,000 articles collected in total.

TABLE 2—DESCRIPTIVE STATISTICS

Indicators	N	Mean	Median	P25	P75	Std Dev
<i>Panel A: Photo and Text Sentiment Indicators for News Samples</i>						
PhotoPes	2982	0.1900	0.1842	0.1419	0.2344	0.0683
TextPes	2982	0.1430	0.1421	0.1000	0.1846	0.0609
<i>Panel B: Returns of Major Chinese Stock Market Indices</i>						
CSI 300	2982	0.0002	0.0003	−0.0058	0.0068	0.0135
SSE Composite	2982	0.0002	0.0005	−0.0050	0.0060	0.0126
SZSE Composite	2982	0.0003	0.0011	−0.0070	0.0086	0.0157
ChiNext Composite	2982	0.0003	−0.0001	−0.0095	0.0103	0.0194
CSI 500	2982	0.0002	0.0009	−0.0067	0.0084	0.0158

Note: The sample period is from 2014 to 2026. The returns of the major Chinese stock market indices are calculated as log daily returns. The full sample contains 2,982 trading days; regression analyses use $N = 2,978$ after accounting for the five-period lag structure. Sentiment indicators are constructed from all news days in the sample.

Figure 3 displays the monthly average trend of the two sentiment indicators, PhotoPes and TextPes. The figure shows that the two sentiment indicators exhibit distinctly different time-series characteristics and volatility patterns between 2014 and 2026. Throughout the sample period, PhotoPes consistently remains at a higher level than TextPes, suggesting that news images generally convey more pessimistic sentiment than textual content. During specific major events, both indicators show significant fluctuations. For instance, both indicators experienced a sharp decline in early 2015, with TextPes reaching a low point of around 0.05. PhotoPes peaked at approximately 0.29 in early 2016 and again at the beginning

of 2018, coinciding with periods of market turbulence. The COVID-19 outbreak in early 2020 was marked by another notable spike in PhotoPes. TextPes showed more pronounced peaks in 2017-2018 during the US-China trade tensions, but has displayed a general downward trend since 2020, reaching its lowest values in late 2024. In contrast, PhotoPes maintained a more volatile pattern throughout the period, typically fluctuating between 0.15 and 0.25, without showing the same consistent downward trend as TextPes in recent years. This divergence suggests that images and text capture different dimensions of market sentiment information, with textual content becoming increasingly neutral over time while images continue to convey stronger emotional signals.

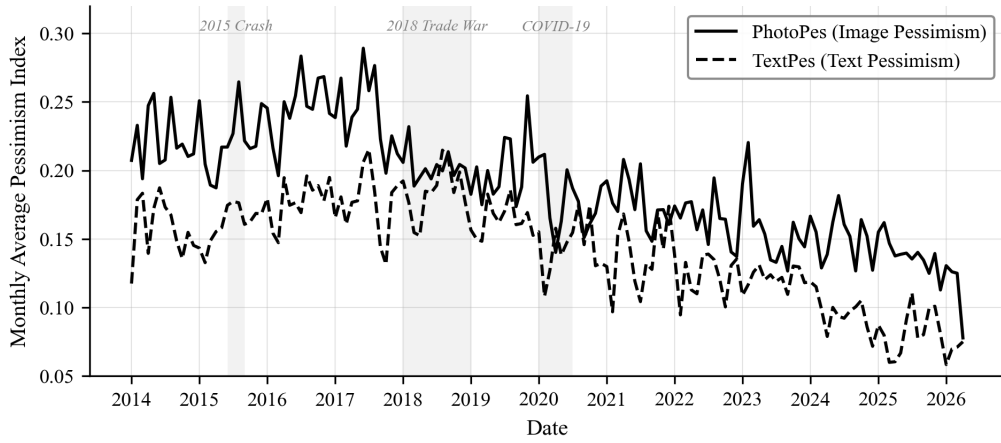


FIGURE 3. TIME-SERIES CHARACTERISTICS OF SENTIMENT INDICATORS

III. Empirical Results

A. The Impact of Photo Sentiment Indicator on Market Returns

Based on sample data from 2014–2026, this paper uses the following model to analyze the impact of news photo pessimistic sentiment, PhotoPes, on market returns:

$$(4) \quad R_t = \beta_1 L5(\text{PhotoPes}_t) + \beta_2 L5(R_t) + \beta_3 L5(R_t^2) + \beta_4 X_t + \epsilon_t$$

where R_t represents the market return rate, using the daily logarithmic returns of five market indices: CSI 300, SSE Composite (SHCOMP), SZSE Composite (SZCOMP), ChiNext Composite, and CSI 500. PhotoPes_t is the photo pessimistic sentiment indicator. To avoid the influence of extreme values on the estimation results, this paper applies a 1% winsorization to this indicator and standardizes

PhotoPes_{*t*}. $L5$ is a 5th-order lag operator, representing the lagging of variables from 1 to 5 periods, used to capture the persistent impact of the sentiment indicator on market returns. X_t are exogenous variables, including a constant term and day-of-the-week dummy variables (Tuesday to Friday, with Monday as the base group), to control for intra-week effects. ϵ_t is the residual term, assumed to be normally distributed and without serial correlation.

Table 3 presents the regression results of PhotoPes on market returns. The results show that PhotoPes_{*t-3*} has a significant negative correlation with market returns. Specifically, when the photo pessimistic sentiment indicator rises, market returns three trading days later tend to fall. This relationship is statistically significant for the ChiNext Composite index (at the 1% level) and for the SZSE Composite (SZCOMP) and CSI 500 indices (at the 10% level).

The results reveal a persistent negative pattern in the medium-lag range. The sum of the coefficients of PhotoPes from $t - 3$ to $t - 5$ is significantly negative for the SZSE Composite (at the 5% level), ChiNext Composite (at the 5% level), and CSI 500 (at the 10% level), suggesting a sustained negative effect over this period. The photo sentiment effect is particularly pronounced in growth-oriented and smaller-cap markets, consistent with the prediction that investor sentiment has a larger impact on stocks with higher information uncertainty and greater difficulty in arbitrage. This finding supports the overreaction hypothesis proposed in behavioral finance by De Bondt and Thaler (1987), which posits that market participants may overreact to negative sentiment information, leading to short-term price deviations from fundamentals, followed by a correction.

From an economic perspective, this effect has a substantial impact. For example, on the ChiNext Composite index, a one-standard-deviation change in PhotoPes leads to a return change on day $t - 3$ of -0.13%, which is economically significant compared to the average daily return of the index (see Table 2).

B. The Impact of Text Sentiment Indicator on Market Returns

To compare the predictive power of the text sentiment indicator with the photo sentiment indicator, this paper employs the same model framework as before to analyze the impact of the text pessimistic sentiment indicator (TextPes) on market returns:

$$(5) \quad R_t = \beta_1 L5(\text{TextPes}_t) + \beta_2 L5(R_t) + \beta_3 L5(R_t^2) + \beta_4 X_t + \epsilon_t$$

where TextPes_{*t*} is the text pessimistic sentiment indicator, constructed based on the articles used for calculating PhotoPes. To avoid the influence of extreme values on the estimation results, this paper applies a 1% winsorization to this indicator and standardizes TextPes_{*t*}.

Table 4 shows the regression results for the 2014–2026 sample period. The results indicate that TextPes_{*t-3*} has a negative correlation with market returns for the SZSE Composite (SZCOMP) index (at the 10% level) and the ChiNext

TABLE 3—REGRESSION RESULTS OF PHOTOPEs ON MARKET RETURNS (2014–2026)

Indicators	<i>CSI 300</i>	<i>SHCOMP</i>	<i>SZCOMP</i>	<i>ChiNext</i>	<i>CSI 500</i>
PhotoPes _{<i>t</i>−1}	0.0000 (0.06)	0.0001 (0.27)	0.0002 (0.56)	0.0000 (0.10)	0.0002 (0.47)
PhotoPes _{<i>t</i>−2}	0.0002 (0.80)	0.0002 (0.92)	0.0004 (1.18)	0.0005 (1.19)	0.0004 (1.11)
PhotoPes _{<i>t</i>−3}	-0.0005 (-1.50)	-0.0004 (-1.19)	-0.0007* (-1.85)	-0.0013*** (-2.81)	-0.0007* (-1.71)
PhotoPes _{<i>t</i>−4}	-0.0003 (-0.94)	-0.0004 (-1.36)	-0.0006* (-1.70)	-0.0006 (-1.48)	-0.0006 (-1.62)
PhotoPes _{<i>t</i>−5}	0.0004 (1.33)	0.0002 (0.84)	0.0004 (1.11)	0.0007* (1.74)	0.0004 (1.10)
Sum(<i>t</i> − 3 to <i>t</i> − 5)	-0.0004 (-0.95)	-0.0005 (-1.39)	-0.0009** (-2.07)	-0.0011** (-1.98)	-0.0009* (-1.94)
Sum(<i>t</i> − 4 to <i>t</i> − 5)	0.0001 (0.32)	-0.0002 (-0.45)	-0.0002 (-0.52)	0.0001 (0.23)	-0.0002 (-0.45)
R ²	0.0151	0.0151	0.0151	0.0150	0.0157
Adj R ²	0.0088	0.0088	0.0088	0.0087	0.0093
N	2978	2978	2978	2978	2978

Note: t-values are in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. Standard errors are Newey-West HAC with 5 lags.

Composite index (at the 10% level). This suggests that after an increase in text pessimistic sentiment, market returns on the third trading day tend to fall, with this effect being particularly pronounced in markets dominated by growth stocks.

The regression results for the TextPes indicator are broadly consistent with the overreaction hypothesis. The coefficient for TextPes_{*t*−3} is negative across most indices, indicating an initial market decline following high text pessimism. The total cumulative effect from *t* − 3 to *t* − 5 is not statistically significant for any index, which suggests the initial negative impact tends to be offset by subsequent market corrections.

From an economic perspective, on the ChiNext Composite index, a one-standard-deviation change in TextPes leads to a return change on day *t* − 3 of approximately −0.07% (see Table 2).

C. Photo Sentiment, Text Sentiment, and Their Interaction Effect

To further investigate the joint impact of the photo sentiment and text sentiment indicators, this paper constructs an extended model that includes both sentiment indicators simultaneously:

$$(6) \quad R_t = \beta_1 L5(\text{PhotoPes}_t) + \beta_2 L5(\text{TextPes}_t) + \beta_3 L5(R_t) + \beta_4 L5(R_t^2) + \beta_5 X_t + \epsilon_t$$

Table 5 shows the regression results for the 2014–2026 sample period. The findings in Panel A indicate that even after controlling for TextPes, the impact

TABLE 4—REGRESSION RESULTS OF TEXTPes ON MARKET RETURNS (2014–2026)

Indicators	<i>CSI 300</i>	<i>SHCOMP</i>	<i>SZCOMP</i>	<i>ChiNext</i>	<i>CSI 500</i>
TextPes _{<i>t</i>−1}	-0.0003 (-0.91)	-0.0003 (-1.10)	-0.0004 (-1.01)	-0.0002 (-0.33)	-0.0004 (-0.93)
TextPes _{<i>t</i>−2}	-0.0001 (-0.31)	-0.0001 (-0.48)	-0.0002 (-0.67)	-0.0006 (-1.32)	-0.0002 (-0.46)
TextPes _{<i>t</i>−3}	-0.0002 (-0.74)	-0.0003 (-1.01)	-0.0005* (-1.66)	-0.0007* (-1.70)	-0.0005 (-1.42)
TextPes _{<i>t</i>−4}	0.0002 (0.67)	0.0002 (0.79)	0.0004 (1.05)	0.0004 (0.86)	0.0003 (0.89)
TextPes _{<i>t</i>−5}	0.0004 (1.07)	0.0003 (0.95)	0.0003 (0.72)	0.0003 (0.67)	0.0001 (0.26)
Sum(<i>t</i> − 3 to <i>t</i> − 5)	0.0003 (0.74)	0.0002 (0.57)	0.0001 (0.20)	0.0000 (0.02)	-0.0001 (-0.12)
Sum(<i>t</i> − 4 to <i>t</i> − 5)	0.0006 (1.29)	0.0005 (1.30)	0.0006 (1.34)	0.0007 (1.17)	0.0004 (0.84)
R ²	0.0145	0.0151	0.0144	0.0127	0.0147
Adj R ²	0.0082	0.0088	0.0080	0.0064	0.0084
N	2978	2978	2978	2978	2978

Note: t-values are in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. Standard errors are Newey-West HAC with 5 lags.

of PhotoPes remains concentrated at the $t - 3$ lag. Specifically, PhotoPes _{$t-3$} continues to exhibit a significant negative predictive effect on market returns for the ChiNext index (at the 5% level), while the effects for SZCOMP and CSI 500 remain directionally consistent but below conventional significance thresholds in the joint specification. This consistency suggests that the photo sentiment indicator captures market sentiment information that is at least partially distinct from textual content.

The text-based pessimistic sentiment indicator (TextPes) in Panel B shows a negative predictive pattern at $t - 3$ that is directionally consistent across all indices. We also estimated a version of this model that additionally includes a $(\text{PhotoPes} \times \text{TextPes})_{t-3}$ interaction term; the interaction coefficient was insignificant at conventional levels across all five indices (untabulated), which supports the view that the contributions of image and text sentiment are largely independent in this sample.

The overall picture from the joint regression reinforces the conclusion that image sentiment—captured through the ViT-based PhotoPes indicator—provides incremental predictive content for Chinese stock market returns beyond what is captured by text sentiment alone.

D. Conditional Analysis: Market Volatility States

Table 6 examines whether the predictive power of sentiment indicators depends on market conditions by splitting the sample into high- and low-volatility

TABLE 5—JOINT EFFECT OF PHOTOPEs AND TEXTPEs ON MARKET RETURNS (2014–2026)

Indicators	<i>CSI 300</i>	<i>SHCOMP</i>	<i>SZCOMP</i>	<i>ChiNext</i>	<i>CSI 500</i>
<i>Panel A: Photo Sentiment Indicator (PhotoPes)</i>					
PhotoPes _{t-1}	0.0000 (0.11)	0.0001 (0.42)	0.0003 (0.81)	0.0001 (0.26)	0.0003 (0.75)
PhotoPes _{t-2}	0.0002 (0.87)	0.0003 (1.11)	0.0005 (1.45)	0.0007 (1.49)	0.0005 (1.38)
PhotoPes _{t-3}	-0.0005 (-1.43)	-0.0003 (-1.05)	-0.0006 (-1.59)	-0.0012** (-2.52)	-0.0006 (-1.45)
PhotoPes _{t-4}	-0.0003 (-1.05)	-0.0004 (-1.38)	-0.0006* (-1.67)	-0.0006 (-1.44)	-0.0006 (-1.52)
PhotoPes _{t-5}	0.0003 (1.13)	0.0002 (0.72)	0.0004 (1.11)	0.0007* (1.71)	0.0004 (1.18)
<i>Panel B: Text Sentiment Indicator (TextPes)</i>					
TextPes _{t-1}	-0.0003 (-0.86)	-0.0003 (-1.12)	-0.0004 (-1.10)	-0.0001 (-0.32)	-0.0004 (-1.03)
TextPes _{t-2}	-0.0001 (-0.29)	-0.0001 (-0.52)	-0.0003 (-0.79)	-0.0006 (-1.41)	-0.0002 (-0.59)
TextPes _{t-3}	-0.0002 (-0.55)	-0.0002 (-0.86)	-0.0005 (-1.51)	-0.0006 (-1.42)	-0.0004 (-1.31)
TextPes _{t-4}	0.0003 (0.79)	0.0003 (0.93)	0.0005 (1.20)	0.0005 (1.03)	0.0004 (1.02)
TextPes _{t-5}	0.0003 (1.03)	0.0003 (0.93)	0.0002 (0.63)	0.0003 (0.58)	0.0001 (0.16)
R ²	0.0162	0.0166	0.0173	0.0169	0.0173
Adj R ²	0.0082	0.0087	0.0093	0.0089	0.0093
N	2978	2978	2978	2978	2978

Note: t-statistics are in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. Standard errors are Newey-West HAC with 5 lags.

states. The volatility state is defined by the 20-day rolling standard deviation of SHCOMP log returns: periods in the top 25th percentile (annualized vol > 19.6%) constitute the high-volatility regime (Panel A, N=741), while the remaining 75% form the low-volatility regime (Panel B, N=2,237).

Panel A reveals that during high-volatility periods, both PhotoPes_{t-3} and TextPes_{t-3} exhibit significant negative predictive power. PhotoPes_{t-3} is significant for the SZSE Composite (10% level), ChiNext (5% level), and CSI 500 (10% level), while TextPes_{t-3} is significant for SZSE Composite, ChiNext, and CSI 500 (all at the 5% level). The coefficient magnitude of PhotoPes_{t-3} is approximately 2.5 times larger in high-volatility periods (−0.0033 for ChiNext) than the full-sample estimate (−0.0013, Table 3), indicating a strong amplification of image-sentiment effects under market stress. The R^2 values are also substantially higher (0.038–0.055) than the full-sample joint regression (0.016–0.017, Table 5), confirming that the model’s explanatory power is concentrated in volatile market conditions.

Panel B shows that, with the sole exception of a marginal TextPes_{t-4} effect on SZCOMP (10% level), neither sentiment indicator is significant at any lag during low-volatility periods. This sharp contrast confirms that the full-sample predictive effects documented in Tables 3 and 4 are driven by high-stress market episodes rather than being uniformly distributed across time.

This state-dependent pattern is consistent with the behavioral finance literature. Loewenstein et al. (2001) argue that emotional processing dominates cognitive processing under stress, causing investors to react more strongly to vivid, emotionally charged stimuli such as news images. Baker and Wurgler (2007) further show that sentiment effects are amplified when arbitrage is most costly—precisely the high-volatility states identified here. The simultaneous significance of both PhotoPes_{t-3} and TextPes_{t-3} during turbulent periods suggests that image and text sentiment jointly amplify investor overreaction, with the return impact materializing three trading days after the initial media signal.

E. Portfolio Construction and Empirical Analysis

In the framework of a sentiment-driven trading strategy, this paper constructs a portfolio based on market sentiment indicators. The portfolio construction process is mainly divided into two stages: signal generation and position allocation. In the signal generation stage, this paper adopts a rule-based approach leveraging the predictive power of *PhotoPes* and *TextPes* sentiment indicators.

This paper first constructs a rolling window of 90 trading days to calculate the rolling quantiles of the combined sentiment signal. Based on previous research findings, we focus particularly on the PhotoPes_{t-3} indicator, which has demonstrated significant negative predictive effects on market returns. The signal generation mechanism identifies extreme sentiment periods by comparing the current combined signal value with its historical 20th and 80th percentile thresholds within the rolling window. When the combined signal is extremely high (above the 80th percentile), a negative market return is anticipated, generating a short signal; conversely, when it is extremely low (below the 20th percentile), a positive market return is expected, generating a long signal.

To enhance the robustness of the signal, this paper also incorporates TextPes_{t-3} as a secondary indicator, with weights of 80% for *PhotoPes* and 20% for *TextPes*, reflecting the stronger in-sample predictive evidence for image sentiment. The combined signal is then processed through a rebalancing mechanism that adjusts positions every 5 trading days to reduce transaction costs and avoid overreacting to short-term market noise.

In the position allocation stage, this paper implements a straightforward approach where the position size is proportional to the strength of the sentiment signal. The strategy maintains a disciplined trading approach, only taking positions when sentiment indicators reach extreme values, which previous research has shown to have the strongest predictive power. This selective approach ensures that the strategy only trades when the probability of success is highest.

TABLE 6—CONDITIONAL IMPACT OF PHOTOPEs AND TEXTPEs BY MARKET VOLATILITY STATE (2014–2026)

Indicators	<i>CSI 300</i>	<i>SHCOMP</i>	<i>SZCOMP</i>	<i>ChiNext</i>	<i>CSI 500</i>
<i>Panel A: High Volatility Periods ($\sigma_t > 75th$ percentile, $N=741$)</i>					
PhotoPes _{<i>t</i>−1}	0.0002 (0.13)	0.0000 (0.03)	0.0012 (0.93)	0.0016 (0.88)	0.0008 (0.63)
PhotoPes _{<i>t</i>−2}	0.0002 (0.25)	0.0005 (0.60)	0.0008 (0.74)	0.0011 (0.85)	0.0008 (0.72)
PhotoPes _{<i>t</i>−3}	−0.0016 (−1.44)	−0.0014 (−1.19)	−0.0021* (−1.73)	−0.0033** (−2.45)	−0.0022* (−1.71)
PhotoPes _{<i>t</i>−4}	−0.0012 (−1.09)	−0.0014 (−1.27)	−0.0017 (−1.38)	−0.0018 (−1.28)	−0.0015 (−1.16)
PhotoPes _{<i>t</i>−5}	0.0007 (0.72)	0.0006 (0.64)	0.0012 (1.03)	0.0016 (1.17)	0.0011 (0.95)
TextPes _{<i>t</i>−1}	−0.0010 (−0.94)	−0.0010 (−1.02)	−0.0011 (−0.95)	−0.0013 (−1.00)	−0.0011 (−0.95)
TextPes _{<i>t</i>−2}	−0.0005 (−0.56)	−0.0005 (−0.57)	−0.0007 (−0.62)	−0.0012 (−0.96)	−0.0002 (−0.18)
TextPes _{<i>t</i>−3}	−0.0015 (−1.48)	−0.0015 (−1.57)	−0.0022** (−2.21)	−0.0026** (−2.34)	−0.0021** (−2.12)
TextPes _{<i>t</i>−4}	0.0004 (0.40)	0.0004 (0.35)	0.0001 (0.11)	0.0002 (0.14)	0.0001 (0.06)
TextPes _{<i>t</i>−5}	0.0017 (1.45)	0.0016 (1.41)	0.0009 (0.71)	0.0010 (0.68)	0.0007 (0.52)
R ²	0.0390	0.0376	0.0446	0.0548	0.0439
<i>Panel B: Low Volatility Periods ($\sigma_t \leq 75th$ percentile, $N=2,237$)</i>					
PhotoPes _{<i>t</i>−1}	−0.0000 (−0.07)	0.0002 (0.87)	0.0001 (0.32)	−0.0003 (−0.76)	0.0002 (0.66)
PhotoPes _{<i>t</i>−2}	0.0002 (0.79)	0.0002 (0.74)	0.0004 (1.20)	0.0004 (1.10)	0.0003 (1.14)
PhotoPes _{<i>t</i>−3}	−0.0001 (−0.52)	−0.0000 (−0.17)	−0.0001 (−0.46)	−0.0005 (−1.25)	−0.0001 (−0.24)
PhotoPes _{<i>t</i>−4}	−0.0001 (−0.42)	−0.0001 (−0.66)	−0.0003 (−1.08)	−0.0003 (−0.95)	−0.0003 (−0.97)
PhotoPes _{<i>t</i>−5}	0.0001 (0.32)	−0.0001 (−0.29)	−0.0000 (−0.13)	0.0003 (0.77)	0.0000 (0.01)
TextPes _{<i>t</i>−1}	−0.0001 (−0.36)	−0.0001 (−0.71)	−0.0002 (−0.82)	0.0001 (0.40)	−0.0002 (−0.81)
TextPes _{<i>t</i>−2}	0.0000 (0.07)	−0.0000 (−0.21)	−0.0002 (−0.61)	−0.0003 (−0.96)	−0.0002 (−0.83)
TextPes _{<i>t</i>−3}	0.0003 (1.46)	0.0002 (1.22)	0.0001 (0.40)	0.0002 (0.42)	0.0002 (0.62)
TextPes _{<i>t</i>−4}	0.0002 (0.68)	0.0002 (0.99)	0.0005* (1.71)	0.0005 (1.35)	0.0004 (1.58)
TextPes _{<i>t</i>−5}	−0.0001 (−0.41)	−0.0002 (−0.78)	0.0000 (0.10)	0.0000 (0.08)	−0.0001 (−0.53)
R ²	0.0118	0.0163	0.0216	0.0157	0.0199

Note: The volatility state is defined by the 20-day rolling standard deviation of SHCOMP log returns; high volatility = top 25th percentile. t-statistics in parentheses. *, **, *** denote significance at 10%, 5%, 1% levels. Standard errors are Newey-West HAC with 5 lags.

To evaluate the effectiveness of the strategy, this paper selects the period from January 2014 to April 2026 as the backtesting period, using the CSI 500 Index as the benchmark. The backtesting process takes into account realistic trading conditions, including a 5-day rebalancing frequency and signal strength-based position sizing.

The empirical results show that the sentiment-driven strategy outperformed the benchmark during the sample period. As shown in Table 7, the strategy achieved an annualized return of 9.03% with an annualized volatility of 24.86%, resulting in a Sharpe ratio of 0.363. In contrast, the CSI 500 Index recorded an annualized return of 6.23% with an annualized volatility of 25.14%, leading to a Sharpe ratio of 0.248. The strategy demonstrated modestly superior risk control; its maximum drawdown was -62.22% , compared to the benchmark’s maximum drawdown of -71.11% .

TABLE 7—PERFORMANCE COMPARISON OF THE STRATEGY AND THE CSI 500 INDEX (2014–2026)

Metrics	Sentiment-Driven Strategy	CSI 500 Index
Annualized Return	9.03%	6.23%
Annualized Volatility	24.86%	25.14%
Sharpe Ratio	0.363	0.248
Maximum Drawdown	-62.22%	-71.11%

Figure 4 illustrates the cumulative returns of the sentiment-driven strategy and the CSI 500 Index. The figure confirms the strategy’s outperformance over the sample period. If 1 RMB were invested at the beginning of 2014, it would have grown to approximately 2.02 RMB by April 2026 under the sentiment-driven strategy, compared to 1.42 RMB under the CSI 500 benchmark.

To further validate the effectiveness of the strategy, this paper conducted a Fama and French (1993) three-factor regression analysis on the strategy’s returns using the CUFE Chinese three-factor data, with the results shown in Table 8. The regression results reveal that the strategy’s alpha is 0.000236 (daily). While positive, this alpha is not statistically significant at conventional levels (p -value of 0.469), which suggests that the return outperformance is not robustly distinguishable from common factor exposure. The HML coefficient is -0.103 and is marginally significant at the 10% level, indicating a weak tilt toward growth-oriented stocks. The very low R^2 of 0.0048 reflects that the strategy’s returns are largely orthogonal to the common three factors.

F. Out-of-Sample Predictive Analysis

This paper employs an expanding-window method for out-of-sample (OOS) predictive analysis to evaluate the predictive ability of sentiment indicators on the returns of Chinese stock market indices. Following the return predictability literature (Campbell and Thompson, 2008), the analysis uses an initial estimation



FIGURE 4. CUMULATIVE RETURNS OF SENTIMENT-DRIVEN STRATEGY AND CSI 500 INDEX (2014–2026)

TABLE 8—THREE-FACTOR REGRESSION ANALYSIS OF THE SENTIMENT-DRIVEN STRATEGY

Factor	Coefficient	<i>t</i> -statistic	<i>p</i> -value
Alpha	0.000236	0.7237	0.4693
MKT	0.0326	0.3723	0.7097
SMB	0.0220	0.3893	0.6970
HML	−0.1026*	−1.7976	0.0722
R^2	0.0048		
Adj. R^2	0.0037		
N	2942		

Note: ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. Fama–French three factors (MKT, SMB, HML) are obtained from the CUFE Chinese Asset Management Research Center database. The regression uses daily excess returns (strategy return minus risk-free rate) over the period 2014-01-06 to 2026-02-06 with Newey–West HAC standard errors (5 lags).

window of 1,512 trading days (approximately six years, covering January 2014 to March 2020) that expands forward one day at a time, yielding 1,467 OOS predictions for each index over the period March 2020 to April 2026. The six-year initial window ensures that the model is trained on a sufficiently diverse set of market conditions—including the 2015 stock market crash, the 2018 trade tensions, and the onset of the COVID-19 pandemic—before generating out-of-sample forecasts. The prediction model uses Ridge regression with a regularization parameter $\alpha = 500$, with standardized *PhotoPes* at lags 3, 4, and 5 as predictors. This multi-lag specification is motivated by the in-sample evidence (Table 3) that both $PhotoPes_{t-3}$ and $PhotoPes_{t-4}$ carry negative predictive information, suggesting that sentiment effects diffuse into prices gradually over several trading days. Ridge regularization penalizes the additional coefficients and prevents over-

fitting, effectively trading off bias for reduced variance in the forecasts (Campbell and Thompson, 2008).

To assess predictive performance, this paper uses the out-of-sample R-squared (R_{OOS}^2) proposed by Campbell and Thompson (2008) and the MSPE-adjusted test statistic proposed by Clark and West (2007):

$$(7) \quad R_{OOS}^2 = 1 - \frac{\sum_{t=1}^T (R_t - \hat{R}_t)^2}{\sum_{t=1}^T (R_t - \bar{R}_t)^2},$$

where R_t is the actual return, $\hat{R}_t$ is the model's predicted return, $\bar{R}_t$ is the expanding-window historical average return, and T is the number of OOS observations. The MSPE-adjusted statistic accounts for the noise introduced by estimating additional parameters and provides a one-sided test of the null hypothesis that the predictive model does not outperform the historical mean.

The OOS results in Table 9 show that all five indices achieve positive R_{OOS}^2 values. The ChiNext Composite attains 0.29%, which is statistically significant at the 5% level based on the MSPE-adjusted test ($t = 2.02$, $p = 0.022$). The SZSE Composite ($R_{OOS}^2 = 0.08\%$, $t = 1.38$, $p = 0.083$) and CSI 500 ($R_{OOS}^2 = 0.07\%$, $t = 1.38$, $p = 0.083$) are both significant at the 10% level. The CSI 300 ($R_{OOS}^2 = 0.04\%$, $t = 1.07$) and SSE Composite ($R_{OOS}^2 = 0.06\%$, $t = 1.22$) are positive with moderate MSPE-adjusted t -statistics, but do not reach conventional significance thresholds, consistent with the weaker in-sample effects for large-cap indices dominated by institutional investors.

Although the magnitudes of R_{OOS}^2 are small in absolute terms, Campbell and Thompson (2008) demonstrate that even modest positive values at the daily frequency can translate into meaningful economic gains for a mean-variance investor. The pattern of OOS predictability—strongest for growth-oriented, smaller-capitalization indices (ChiNext, SZSE Composite, CSI 500) with higher retail investor participation, and weakest for the large-cap CSI 300 and SSE Composite—is consistent with behavioral finance theory that sentiment effects are most pronounced for stocks that are harder to value and more difficult to arbitrage (Baker and Wurgler, 2007). The rule-based sentiment strategy described in the portfolio construction subsection operates on pre-identified signal thresholds rather than estimated regression coefficients, and its performance should be interpreted independently from the OOS regression results.

IV. Robustness Checks

To verify the robustness of the predictive power of the photo sentiment indicator (PhotoPes) on market returns, this paper conducted a series of robustness checks. Table 10 reports the regression results under three different settings.

Considering that winsorization might affect the information content of the sentiment indicator, this paper constructed a PhotoPes indicator without winsorization for analysis. As shown in Panel A of Table 10, the key finding holds:

TABLE 9—OUT-OF-SAMPLE PREDICTIVE PERFORMANCE OF SENTIMENT INDICATORS

Indicators	<i>CSI 300</i>	<i>SHCOMP</i>	<i>SZCOMP</i>	<i>ChiNext</i>	<i>CSI 500</i>
R_{OOS}^2 (%)	0.04	0.06	0.08*	0.29**	0.07*
MSPE-adj t -stat	1.07	1.22	1.38	2.02	1.38

Note: The prediction model is an expanding-window Ridge regression ($\alpha = 500$) with standardized $PhotoPes_{t-3}$, $PhotoPes_{t-4}$, and $PhotoPes_{t-5}$ as predictors. The initial estimation window is 1,512 trading days (approximately six years). The OOS period runs from March 2020 to April 2026, yielding 1,467 predictions. R_{OOS}^2 values are reported in percentage terms. ***, **, and * indicate significance at the 1%, 5%, and 10% levels based on the one-sided MSPE-adjusted test of Clark and West (2007).

$PhotoPes_{t-3}$ shows a significant negative coefficient in the SZSE Composite (SZCOMP) at the 10% level, ChiNext at the 1% level, and CSI 500 at the 10% level. This indicates that the main findings are not driven by the winsorization procedure.

Cox and Peterson (1994) argue that extreme price returns are often accompanied by short-term price reversals. To eliminate the potential impact of high market volatility periods on the results, this paper uses GARCH(1,1)-adjusted returns as the dependent variable. As shown in Panel B of Table 10, after adjusting for volatility, $PhotoPes_{t-3}$ retains a significant negative coefficient in the ChiNext index at the 5% level, while the effect is attenuated for other indices, confirming the robustness of the core finding for growth-oriented markets under volatility controls.

To exclude the potential bias caused by extreme market returns, this paper removes the most extreme returns from the sample (0.5% from each tail of SHCOMP). As shown in Panel C of Table 10, the predictive power of $PhotoPes_{t-3}$ remains robust for the ChiNext index at the 5% level. This confirms that the indicator’s core predictive ability is not solely driven by extreme market events.

Under these three different settings, the significant negative coefficient of $PhotoPes_{t-3}$ remains consistent for the ChiNext index, which represents growth-oriented and small-to-mid-cap stocks. This series of checks confirms that the negative predictive power of the photo sentiment indicator is a robust phenomenon, particularly pronounced in markets where investor sentiment plays a more prominent role in asset pricing.

V. Conclusion

This paper, using textual and visual information from news media as a starting point, explores methods for constructing investor sentiment indicators based on multimodal data and empirically tests their effectiveness in the Chinese stock market. By introducing deep learning techniques, the paper constructs a photo-based pessimism indicator, PhotoPes, from news images and a text-based pessimism indicator, TextPes, from news texts. It then conducts empirical regressions and investment strategy analysis using various market indices to uncover the mecha-

TABLE 10—IMPACT OF PHOTOPEs ON MARKET RETURNS - ROBUSTNESS CHECKS (2014–2026)

Indicators	<i>CSI 300</i>	<i>SHCOMP</i>	<i>SZCOMP</i>	<i>ChiNext</i>	<i>CSI 500</i>
<i>Panel A: Without Winsorization</i>					
PhotoPes _{<i>t</i>−1}	0.0021 (0.46)	0.0010 (0.21)	0.0042 (0.75)	0.0015 (0.20)	0.0040 (0.71)
PhotoPes _{<i>t</i>−2}	0.0023 (0.60)	0.0025 (0.60)	0.0046 (0.88)	0.0067 (1.02)	0.0040 (0.77)
PhotoPes _{<i>t</i>−3}	-0.0054 (-1.22)	-0.0071 (-1.52)	-0.0097* (-1.82)	-0.0175*** (-2.79)	-0.0093* (-1.70)
PhotoPes _{<i>t</i>−4}	-0.0055 (-1.38)	-0.0042 (-1.00)	-0.0081 (-1.62)	-0.0083 (-1.40)	-0.0080 (-1.58)
PhotoPes _{<i>t</i>−5}	0.0035 (0.86)	0.0059 (1.33)	0.0053 (1.05)	0.0100 (1.56)	0.0054 (1.07)
R ²	0.0160	0.0162	0.0156	0.0152	0.0164
Adj R ²	0.0096	0.0098	0.0092	0.0088	0.0100
N	2944	2944	2944	2944	2944
<i>Panel B: Adjusting for Volatility</i>					
PhotoPes _{<i>t</i>−1}	0.1896 (0.60)	0.0063 (0.02)	0.1700 (0.53)	0.0246 (0.08)	0.1422 (0.45)
PhotoPes _{<i>t</i>−2}	0.0766 (0.26)	0.1427 (0.47)	0.2433 (0.76)	0.2823 (0.87)	0.1869 (0.60)
PhotoPes _{<i>t</i>−3}	-0.2233 (-0.68)	-0.2515 (-0.77)	-0.3858 (-1.20)	-0.6382** (-1.99)	-0.3523 (-1.10)
PhotoPes _{<i>t</i>−4}	-0.1915 (-0.66)	-0.0602 (-0.20)	-0.3347 (-1.12)	-0.3281 (-1.10)	-0.3190 (-1.07)
PhotoPes _{<i>t</i>−5}	0.1978 (0.68)	0.3385 (1.14)	0.2294 (0.78)	0.3378 (1.14)	0.2514 (0.86)
R ²	0.0102	0.0096	0.0115	0.0088	0.0107
Adj R ²	0.0038	0.0032	0.0051	0.0024	0.0043
N	2944	2944	2944	2944	2944
<i>Panel C: Removing Extreme Market Returns</i>					
PhotoPes _{<i>t</i>−1}	0.0059* (1.76)	0.0046 (1.29)	0.0084* (1.93)	0.0065 (1.19)	0.0079* (1.78)
PhotoPes _{<i>t</i>−2}	0.0065** (2.01)	0.0070** (1.97)	0.0075* (1.66)	0.0082 (1.40)	0.0071 (1.58)
PhotoPes _{<i>t</i>−3}	-0.0039 (-1.06)	-0.0050 (-1.23)	-0.0072 (-1.56)	-0.0130** (-2.27)	-0.0061 (-1.34)
PhotoPes _{<i>t</i>−4}	-0.0035 (-0.97)	-0.0026 (-0.67)	-0.0066 (-1.44)	-0.0076 (-1.37)	-0.0052 (-1.12)
PhotoPes _{<i>t</i>−5}	-0.0038 (-1.16)	-0.0017 (-0.47)	-0.0026 (-0.57)	0.0016 (0.27)	-0.0034 (-0.76)
R ²	0.0207	0.0176	0.0166	0.0117	0.0173
Adj R ²	0.0143	0.0111	0.0101	0.0052	0.0109
N	2914	2914	2914	2914	2914

Note: This table reports robustness checks for the impact of PhotoPes on market returns. Panel A uses PhotoPes without winsorization. Panel B uses volatility-adjusted returns. Panel C removes extreme market returns (top and bottom 0.5%). t-statistics are reported in parentheses. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively.

nisms by which visual and textual sentiment affect investor behavior and market performance.

The research findings show that photo sentiment has significant impact on market returns over the 2014–2026 sample period. Specifically, $PhotoPes_{t-3}$ demonstrates consistent negative predictive power, with particularly strong effects in growth-oriented markets such as ChiNext (significant at the 1% level) and smaller-cap indices. This finding is directionally consistent with the overreaction hypothesis in behavioral finance and persists in the joint regression specification controlling for text sentiment.

Text sentiment ($TextPes_{t-3}$) shows complementary predictive patterns, particularly for the SZSE Composite (SZCOMP) and ChiNext indices, though with somewhat weaker statistical significance in this extended sample. The conditional analysis further shows that these predictive effects are concentrated in high-volatility market states: during high-volatility periods, both $PhotoPes_{t-3}$ and $TextPes_{t-3}$ are significant predictors of returns on the SZSE Composite, ChiNext, and CSI 500, whereas neither indicator exhibits meaningful predictive power during low-volatility periods.

In terms of investment applications, this paper constructs a rule-based trading strategy that leverages the predictive power of $PhotoPes_{t-3}$ and $TextPes_{t-3}$. The strategy achieves positive risk-adjusted returns, outperforming the CSI 500 benchmark with a Sharpe ratio of 0.363 versus 0.248 for the benchmark. A three-factor regression analysis using CUFE factors shows the strategy’s alpha is positive (0.000236 daily) but does not achieve statistical significance at conventional levels, suggesting that future work should explore whether the strategy performance can be made more robust to the factor model.

Although this paper has conducted systematic explorations in methodology and application, it still has certain limitations. First, while the image and text sentiment recognition models used are based on the current mainstream ViT and BERT architectures, the generalization ability of the models is still limited by the distribution and quality of the training corpora. Future research could further optimize these models by introducing professional image and text corpora specific to the financial context. Second, the construction of sentiment indicators is at a daily market granularity, which does not capture higher-frequency sentiment changes and thus cannot meet the needs of high-frequency trading or event-driven strategies. Additionally, this paper mainly focuses on the predictive role of sentiment variables on market returns but does not delve into the behavioral pathways through which sentiment affects the market, such as the heterogeneity of investor types and information dissemination mechanisms. Future research could further expand on micro-mechanism modeling and causal identification.

Looking ahead, with the development of large multimodal models and the continuous enrichment of financial data, the analysis of image and text sentiment will play an increasingly important role in the financial field. Researchers can further integrate image and text sentiment with multi-source heterogeneous data

such as social media, search behavior, and trading account data to construct a more detailed and dynamic market sentiment map. At the same time, sentiment indicators can also be embedded in asset pricing, portfolio management, and financial regulation frameworks, becoming an important bridge connecting investment behavior and market mechanisms.

REFERENCES

- Antweiler, Werner, and Murray Z Frank.** 2004. “Is all that talk just noise? The information content of internet stock message boards.” *The Journal of finance*, 59(3): 1259–1294.
- Baker, Malcolm, and Jeffrey Wurgler.** 2006. “Investor sentiment and the cross-section of stock returns.” *The journal of Finance*, 61(4): 1645–1680.
- Baker, Malcolm, and Jeffrey Wurgler.** 2007. “Investor sentiment in the stock market.” *Journal of economic perspectives*, 21(2): 129–151.
- Bollen, Johan, Huina Mao, and Xiaojun Zeng.** 2011. “Twitter mood predicts the stock market.” *Journal of computational science*, 2(1): 1–8.
- Bu, Hui, Zheng Xie, Jiahong Li, and Junjie Wu.** 2018. “Investor sentiment extracted from internet stock message boards and its effect on Chinese stock market.” *Journal of Management Sciences in China*, 21(04): 86–101.
- Campbell, John Y, and Samuel B Thompson.** 2008. “Predicting excess stock returns out of sample: Can anything beat the historical average?” *The Review of Financial Studies*, 21(4): 1509–1531.
- Chen, Hailiang, Prabuddha De, Yu Hu, and Byoung-Hyoun Hwang.** 2014. “Wisdom of crowds: The value of stock opinions transmitted through social media.” *The review of financial studies*, 27(5): 1367–1403.
- Clark, Todd E, and Kenneth D West.** 2007. “Approximately normal tests for equal predictive accuracy in nested models.” *Journal of econometrics*, 138(1): 291–311.
- Cox, Don R, and David R Peterson.** 1994. “Stock returns following large one-day declines: Evidence on short-term reversals and longer-term performance.” *The Journal of Finance*, 49(1): 255–267.
- Cui, Yiming, Wanxiang Che, Ting Liu, Bing Qin, Shijin Wang, and Guoping Hu.** 2020. “Revisiting pre-trained models for Chinese natural language processing.” *arXiv preprint arXiv:2004.13922*.
- Da, Zhi, Joseph Engelberg, and Pengjie Gao.** 2011. “In search of attention.” *The journal of finance*, 66(5): 1461–1499.

- De Bondt, Werner FM, and Richard H Thaler.** 1987. "Further evidence on investor overreaction and stock market seasonality." *The Journal of finance*, 42(3): 557–581.
- De Long, J Bradford, Andrei Shleifer, Lawrence H Summers, and Robert J Waldmann.** 1990. "Noise trader risk in financial markets." *Journal of political Economy*, 98(4): 703–738.
- Devlin, Jacob, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova.** 2018. "BERT: Pre-training of deep bidirectional transformers for language understanding." *arXiv preprint arXiv:1810.04805*.
- Dosovitskiy, Alexey, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al.** 2020. "An image is worth 16x16 words: Transformers for image recognition at scale." *arXiv preprint arXiv:2010.11929*.
- Fama, Eugene F, and Kenneth R French.** 1993. "Common risk factors in the returns on stocks and bonds." *Journal of financial economics*, 33(1): 3–56.
- Foucault, Thierry, David Sraer, and David J Thesmar.** 2011. "Individual investors and volatility." *The Journal of Finance*, 66(4): 1369–1406.
- Garcia, M., and P. Stark.** 1991. *Eyes On the News*. St. Petersburg, FL: Poynter Institute for Media Studies.
- Kahneman, Daniel.** 2011. *Thinking, fast and slow*. Farrar, Straus and Giroux.
- Kraus, Mathias, and Stefan Feuerriegel.** 2017. "Decision support from financial disclosures with deep neural networks and transfer learning." *Decision Support Systems*, 104: 38–48.
- Kumar, Alok, and Charles MC Lee.** 2006. "Retail investor sentiment and return comovements." *The Journal of Finance*, 61(5): 2451–2486.
- Lee, Charles MC, Andrei Shleifer, and Richard H Thaler.** 1991. "Investor sentiment and the closed-end fund puzzle." *The journal of finance*, 46(1): 75–109.
- Lindquist, Kristen A, Tor D Wager, Hedy Kober, Eliza Bliss-Moreau, and Lisa Feldman Barrett.** 2012. "The brain basis of emotion: a meta-analytic review." *Behavioral and brain sciences*, 35(3): 121–143.
- Liu, Weiqi, and Xinxin Liu.** 2014. "Individual /institutional investor sentiment and stock returns: Study based on Shanghai A-share market." *Journal of Management Sciences in China*, 17(03): 70–87.

- Loewenstein, George F, Elke U Weber, Christopher K Hsee, and Ned Welch.** 2001. "Risk as feelings." *Psychological Bulletin*, 127(2): 267–286.
- Loughran, Tim, and Bill McDonald.** 2011. "When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks." *The Journal of Finance*, 66(1): 35–65.
- Meng, Xuejing, Xianglan Meng, and Yangyang Hu.** 2016. "Research on Investor Sentiment Index Based on Text Mining and Baidu Index." *Macroecomics*, 2016(01): 144–153.
- Obaid, Khaled, and Kuntara Pukthuanthong.** 2022. "A picture is worth a thousand words: Measuring investor sentiment by combining machine learning and photos from news." *Journal of Financial Economics*, 144(1): 273–297.
- Pang, Bo, Lillian Lee, et al.** 2008. "Opinion mining and sentiment analysis." *Foundations and Trends® in information retrieval*, 2(1–2): 1–135.
- Peng, Yangtuo, and Hui Jiang.** 2015. "Leverage financial news to predict stock price movements using word embeddings and deep neural networks." *arXiv preprint arXiv:1506.07220*.
- Powell, Thomas E, Hajo G Boomgaarden, Knut De Swert, and Claes H de Vreese.** 2015. "A clearer picture: The contribution of visuals and text to framing effects." *Journal of Communication*, 65(6): 997–1017.
- Shleifer, Andrei, and Robert W Vishny.** 1997. "The limits of arbitrage." *The Journal of finance*, 52(1): 35–55.
- Tetlock, Paul C.** 2007. "Giving content to investor sentiment: The role of media in the stock market." *The Journal of finance*, 62(3): 1139–1168.
- Wang, Changyun, and Jiawei Wu.** 2015. "Media Tone, Investor Sentiment and IPO Pricing." *Journal of Financial Research*, 2015(09): 174–189.
- Wen, Fenghua, Longhao Xu, Guangda Ouyang, and Gang Kou.** 2019. "Retail investor attention and stock price crash risk: evidence from China." *International Review of Financial Analysis*, 65: 101376.
- Yang, Yi, Mark Christopher Siy Uy, and Allen Huang.** 2020. "Finbert: A pretrained language model for financial communications." *arXiv preprint arXiv:2006.08097*.
- You, Quanzeng, Jiebo Luo, Hailin Jin, and Jianchao Yang.** 2015. "Robust image sentiment analysis using progressively trained and domain transferred deep networks." Vol. 29.

Zhang, Jiaxing, Ruyi Gan, Junjie Wang, Yuxiang Zhang, Lin Zhang, Ping Yang, Xinyu Gao, Ziwei Wu, Xiaoqun Dong, Junqing He, Jianheng Zhuo, Qi Yang, Yongfeng Huang, Xiayu Li, Yanghan Wu, Junyu Lu, Xinyu Zhu, Weifeng Chen, Ting Han, Kunhao Pan, Rui Wang, Hao Wang, Xiaojun Wu, Zhongshen Zeng, and Chongpei Chen. 2022. “Fengshenbang 1.0: Being the Foundation of Chinese Cognitive Intelligence.” *CoRR*, abs/2209.02970.

Zhang, Zongxin, and Zhaoying Wu. 2021. “Media’s Emotional Contagion and Analyst Optimistic Bias: Evidence Based on the Technique of Machine Learning.” *Journal of Management World*, 37(01): 170–185+11+20–22.

EXAMPLES OF NEWS PHOTOS WITH EXTREME PESSIMISTIC SENTIMENT

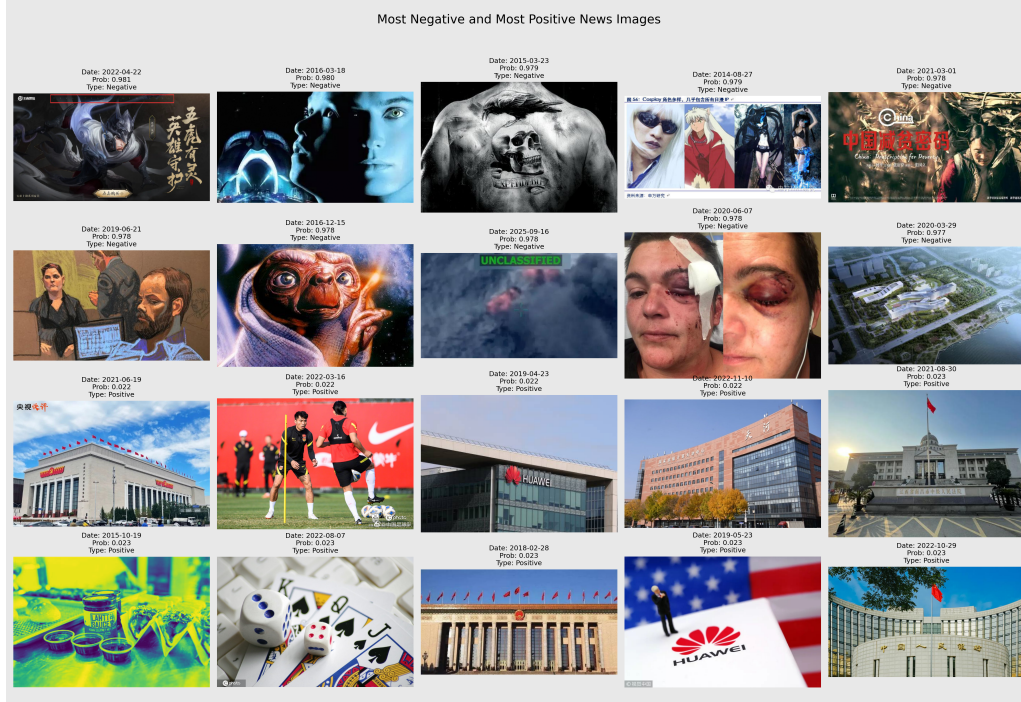


FIGURE A1. EXAMPLES OF NEWS PHOTOS WITH EXTREME PESSIMISTIC SENTIMENT (PHOTOPEs SCORES AT THE 10% TAILS OF THE SAMPLE)